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CNN-Based Phase Imaging: A Quantitative Performance Analysis

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Introduction

Imaging live cells presents a challenge in optical microscopy since they are nearly transparent and invisible under a conventional microscope. While staining can overcome this limitation, it is disruptive to the cells, changing their behavior or even killing them, and is therefore may not always be an appropriate method for imaging dynamic processes. Quantitative phase imaging (QPI) is a class of techniques to render the phase (or delay of light through a cell) visible, which allows higher-contrast imaging of live cells as well as quantitative estimates of cell volume. Convolutional neural networks (CNN) can be used as an alternative method for obtaining phase from defocused images. This research studies the impact of the architectures of CNNs on their performance.

Background

- A specimen can absorb light, changing the intensity, I , measured by a camera focused on it.
- The delay of light through matter can be quantified through a phase delay ϕ , but phase cannot be measured directly since light oscillates too fast.
- The transport of intensity equation (TIE) is a partial differential equation that relates measured intensity and phase.
- For a weakly absorbing sample, such as a live cell, if the intensity in focus is I and the wavelength of light is λ , the variation of intensity on either side of focus at a distance Δz is related to the Laplacian of the object's phase by the transport of intensity equation (TIE):

$$\nabla_{x,y}^2 \phi(x, y, z) = -\frac{2\pi I(x, y, z + \Delta z) - I(x, y, z - \Delta z)}{\lambda I} \quad (2)$$

- The TIE can be solved computationally using standard methods, but these can introduce artifacts due to noise, system misalignments, or the assumed boundary conditions.
- We have previously found that by using high-quality, artifact-free reconstructions of cells, and training a convolutional neural network (CNN) to match the right-hand-side of Eq. (2) to reconstructed phase, the CNN can produce relatively artifact-free reconstructions [1].
- This CNN was trained using 49,328 training images with a 12-hour training time and a fixed architecture.
- In this work we study the performance of networks with reduced training samples and varying architecture.

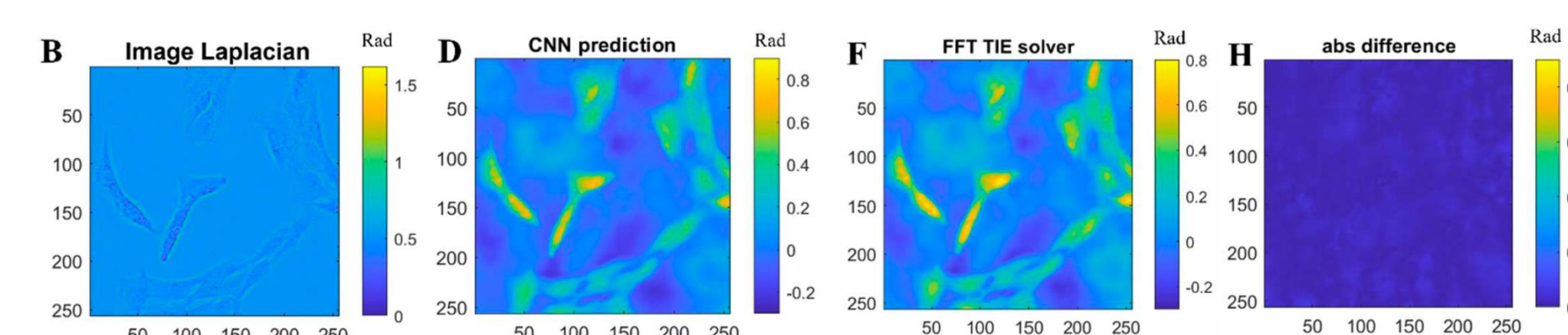


Figure 1. Comparing the CNN's accuracy with the FFT TIE solver (ground truth)

Materials and Methods

Study Design:

We vary two important hyperparameters in CNNs:

- The number of neural filters, small matrices of weights that allow models to extract patterns
- The number of blocks, series of neural layers that determine the hierarchical complexity of models.
- We also evaluated the impact of the training data size. The architecture of these CNNs is designed as follows:

```
def build_UNET_model(self, filters, input_shape, blocks):
    X_input = Input(input_shape)
    convs = [X_input]
    for i in range(blocks):
        conv = Conv2D(filters * (2 ** i), (3, 3), activation="relu", padding="same")(X)
        conv = Conv2D(filters * (2 ** i), (3, 3), activation="relu", padding="same")(conv)
        X = Dropout(0.25 if i == 0 else 0.5)(MaxPooling2D((2, 2))(conv))
        convs.append(conv)
    X = Conv2D(filters * (2 ** blocks), (3, 3), activation="relu", padding="same")(X)
    X = Conv2D(filters * (2 ** blocks), (3, 3), activation="relu", padding="same")(X)
    for i in reversed(range(blocks)):
        X = Conv2DTranspose(filters * (2 ** i), (3, 3), strides=(2, 2), padding="same")(X)
        X = concatenate([X, convs[i]], axis=3)
        X = Conv2D(filters * (2 ** i), (3, 3), activation="relu", padding="same")(Dropout(0.5)(X))
        X = Conv2D(filters * (2 ** i), (3, 3), activation="relu", padding="same")(X)
    return Model(inputs=X_input, outputs=Conv2D(1, 1, activation="sigmoid")(X), name="UNET_dynamic")
```

Figure 2. Code block of build_UNET_model given filters, input_shape, and blocks

The CNN model is trained with a batch size proportional to the number of training samples and for a total of 300 epochs, with a mean-squared-error loss function between the reconstructed phase and the true, high-quality, phase reconstructed from standard methods.

- All three of these studies monitored the MSE and SSIM per epoch during training periods.
- We used a learning rate of 0.0001 and the Adam optimizer in TensorFlow's Keras library.
- Additionally, we performed the training on Google Colab's A100 GPU and conducted three studies A, B, and C.

Study A:

In study A, CNNs were fixed under 4 blocks and 1024 training samples. The four CNNs varied in the number of neural filters (16, 32, 64, 128).

Study B:

In study B, CNNs were fixed under 64 neural filters and 1024 training samples. The four CNNs varied in the number of blocks of layers (2, 3, 4, 5).

Study C:

In study C, CNNs were fixed under 64 neural filters and 4 blocks. The four CNNs were trained under different sample sizes (128, 256, 512, 1024)

References

- Sheikhsofla, et al., 2024 International Conference on Electrical, Computer and Energy Technologies (ICECET, Sydney, Australia, 2024, pp. 1-7, doi: 10.1109/ICECET61485.2024.10698599.
- Wang, K., Song, L., Wang, C. et al. On the use of deep learning for phase recovery. Light Sci Appl 13, 4 (2024). <https://doi.org/10.1038/s41377-023-01340-x>
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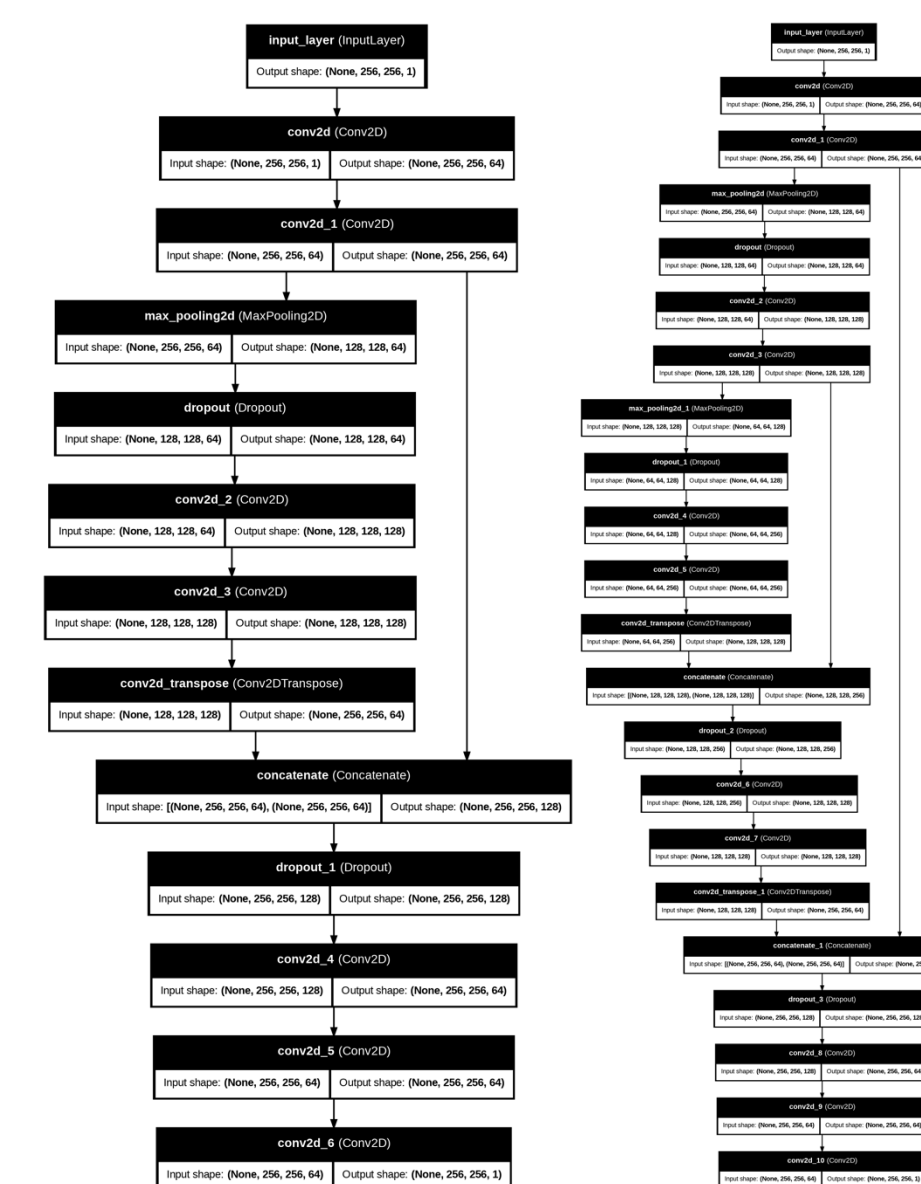


Figure 3. Visual of network architecture when blocks is set to 1 (left) and 2 (right)

Results

- Based on **Study A**, we can see in Figure 4a that models with larger number of filters tend to have lower MSE.
- As more filters are added, the models can detect more distinct patterns.
- This can be also explained in Figure 4b, as predictions from models with more filters are closer to ground truth images. However, a larger number of filters comes with longer training times.
- Additionally, the CNN model with 128 filters seems to have minimal reduction in MSE based on the validation data compared to that with 64 filters.

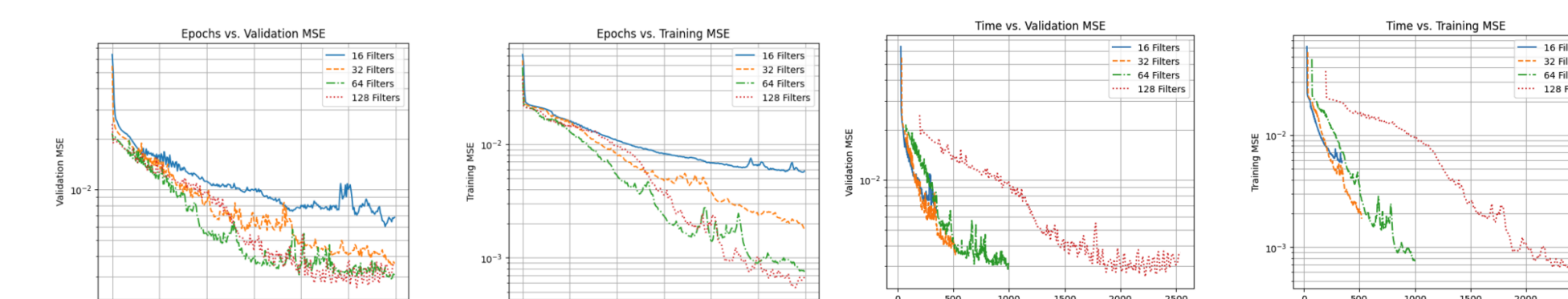


Figure 4a

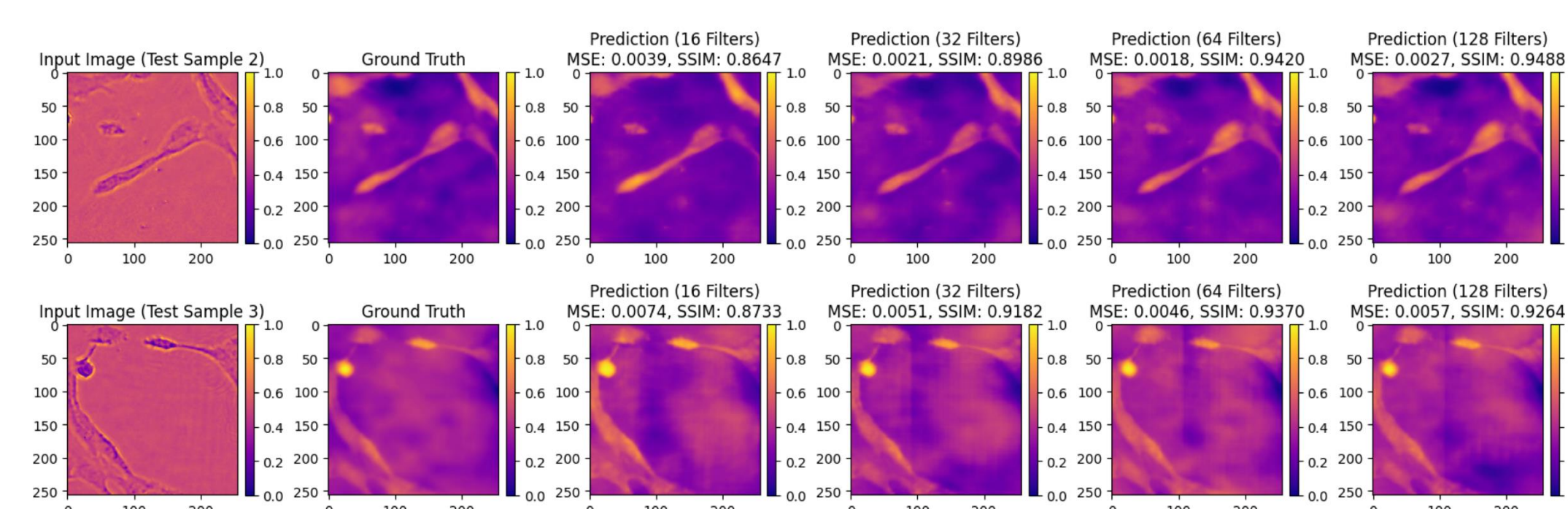


Figure 4b

- In **Study B**, as Figure 5a shows, models with a greater number of layers generally achieve lower MSE, as adding layers enables the model to capture more complex features.
- This is further illustrated in Figure 5b, where predictions from deeper models more closely resemble the ground truth images.
- However, increasing the number of layers is not always necessary due to the relative simplicity of detecting phase patterns.
- The CNN model with 5 blocks shows little to no improvement in MSE compared to the model with 4 blocks, indicating diminishing returns with additional layers.

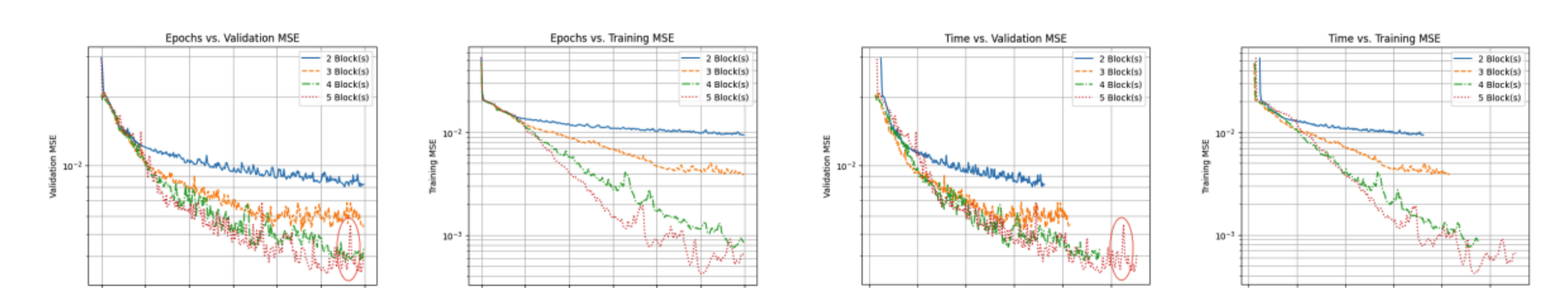


Figure 5a

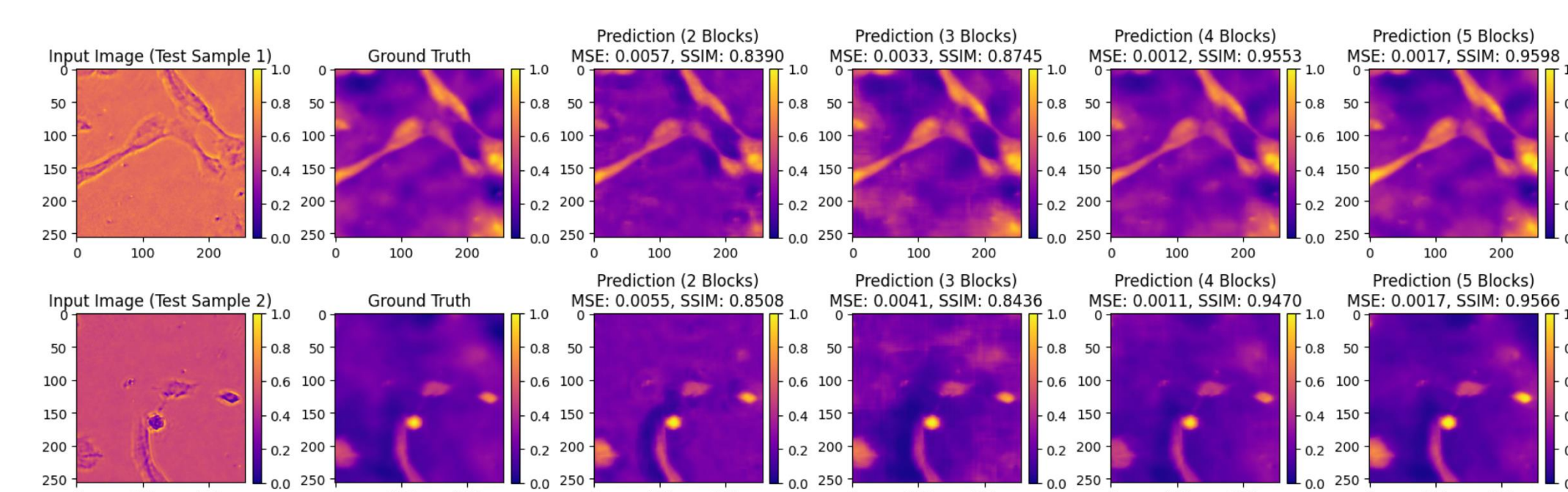


Figure 5b

- Based on **Study C**, Figure 6a shows that models with a larger number of training samples generally achieve lower MSE, as larger sample sizes allow the model to learn from a wider variety of patterns, similar to the effect of increasing filters.
- This is further illustrated in Figure 6b, where predictions from models trained with more samples are closer to the ground truth images.
- Increasing the sample size consistently improves prediction accuracy. Nevertheless, larger sample sizes come at the cost of longer training times.

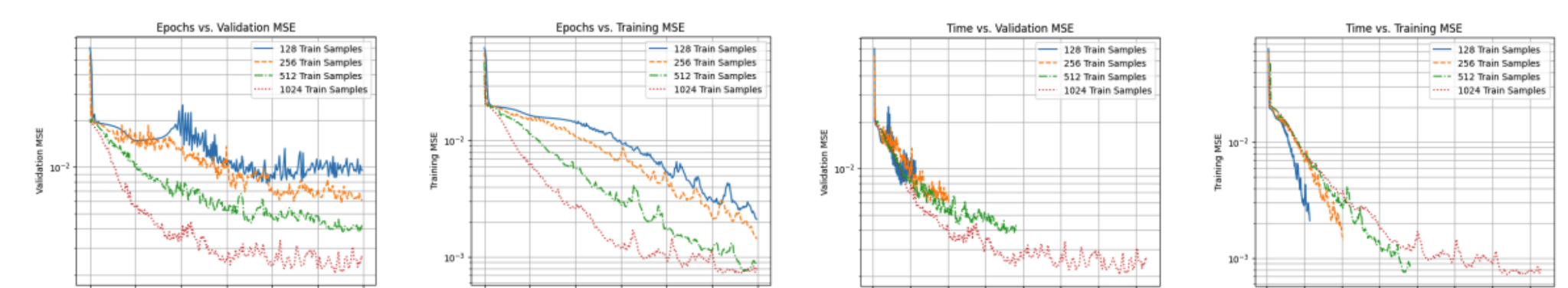


Figure 6a

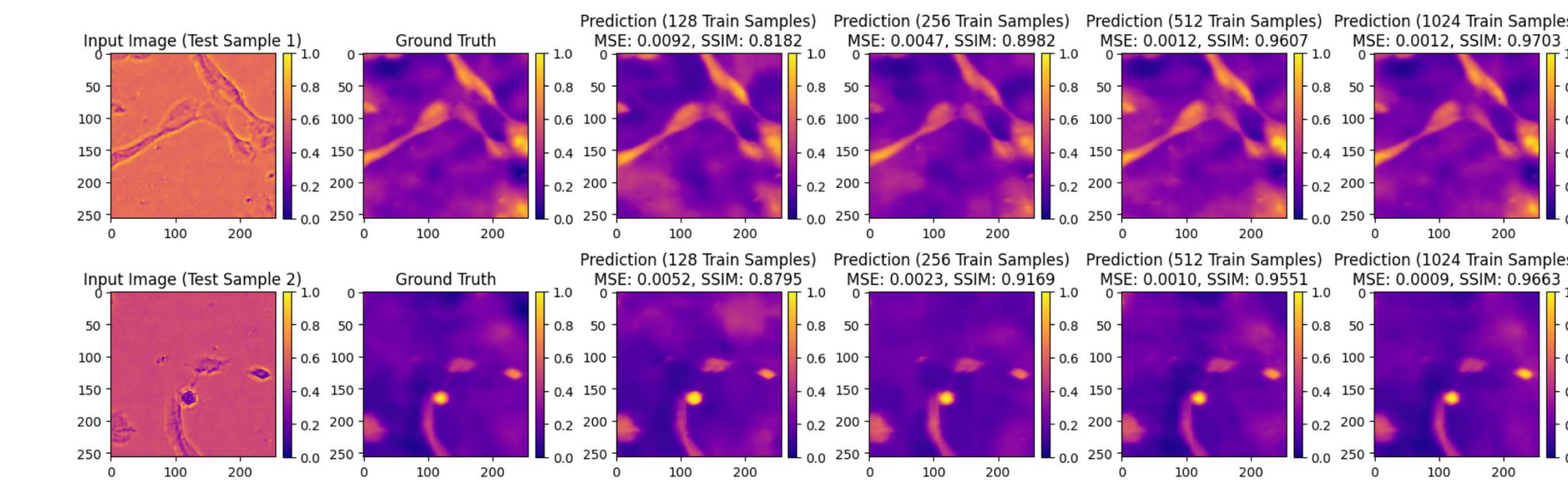


Figure 6b

Conclusion

Our quantitative analysis demonstrates that CNN-based phase imaging can achieve high accuracy, even with a small number of training samples. Models with more filters tend to perform better as they enhance pattern recognition. Similarly, models with additional layers can improve accuracy by detecting more complex patterns, though an excessive number of layers can reduce learning efficiency. Increasing the amount of training data also boosts accuracy by enabling better pattern recognition. However, higher prediction accuracy usually requires longer training times.